

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 6

Total Marks:30

Annexure A

Scenario-Based

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Ragul

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Pharmacy Management System** enables customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage medicine inventory and user accounts. The interactions among system users are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

Answer:

Actors

- Customer
- Pharmacist
- Administrator
- Payment Gateway (External System)

Use Cases

Customer

- Register/Login
- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Track Order

Pharmacist

- Verify Prescription
- Dispense Medicines
- Update Order Status

Administrator

- Manage Medicine Inventory
- Manage User Accounts
- Generate Reports

Payment Gateway

- Process Payment

Relationships

<<include>>

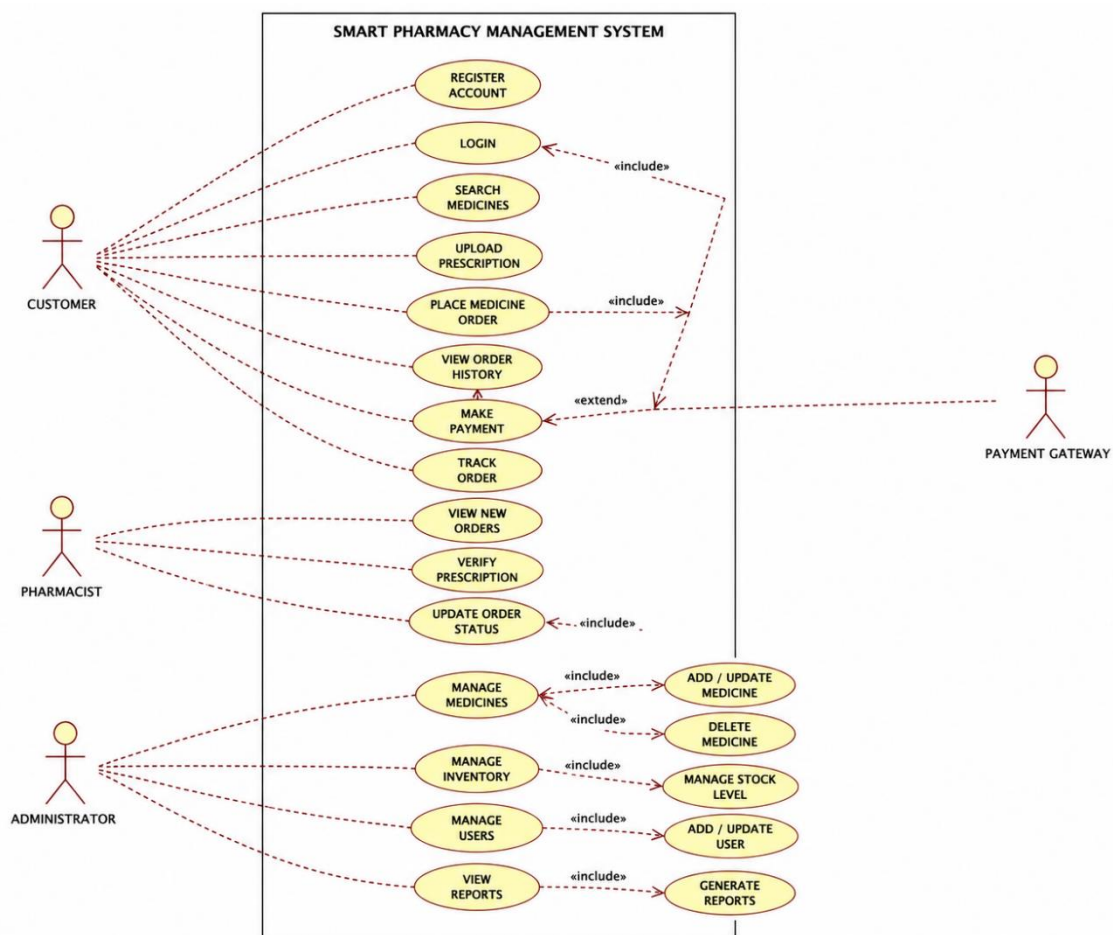
- Place Medicine Order → Search Medicines
- Place Medicine Order → Upload Prescription
- Make Online Payment → Process Payment

<<extend>>

- Verify Prescription → Place Medicine Order
- Track Order → Place Medicine Order

Generalization

- User
 - Customer
 - Pharmacist
 - Administrator



2. Class Diagram Scenario

A **Smart Hotel Management System** consists of entities such as Guest, Room, Reservation, Payment, and Receptionist. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

Answer:

Classes

1. Guest

Attributes

- guestId : int
- name : String
- phone : String
- email : String

Methods

- register()
- bookRoom()
- cancelReservation()

2. Room

Attributes

- roomId : int
- roomType : String
- roomPrice : double
- status : String

Methods

- checkAvailability()
- updateStatus()

3. Reservation

Attributes

- reservationId : int

- checkInDate : Date
- checkOutDate : Date
- status : String

Methods

- createReservation()
- cancelReservation()
- confirmReservation()

4. Payment

Attributes

- paymentId : int
- amount : double
- paymentMethod : String
- paymentStatus : String

Methods

- makePayment()
- generateReceipt()

5. Receptionist

Attributes

- receptionistId : int
- name : String
- shift : String

Methods

- assignRoom()
- checkInGuest()
- checkOutGuest()

6. Person (Superclass)

Attributes

- id : int
- name : String

Methods

- login()
- logout()

Relationships

Inheritance

- Person <|-- Guest
- Person <|-- Receptionist

Association

- Guest ↔ Reservation (**1 : 0..***)
- Receptionist ↔ Reservation (**1 : 0..***)

Aggregation

- Reservation ◇— Room (**1 : 1**)
- One reservation is associated with one room.

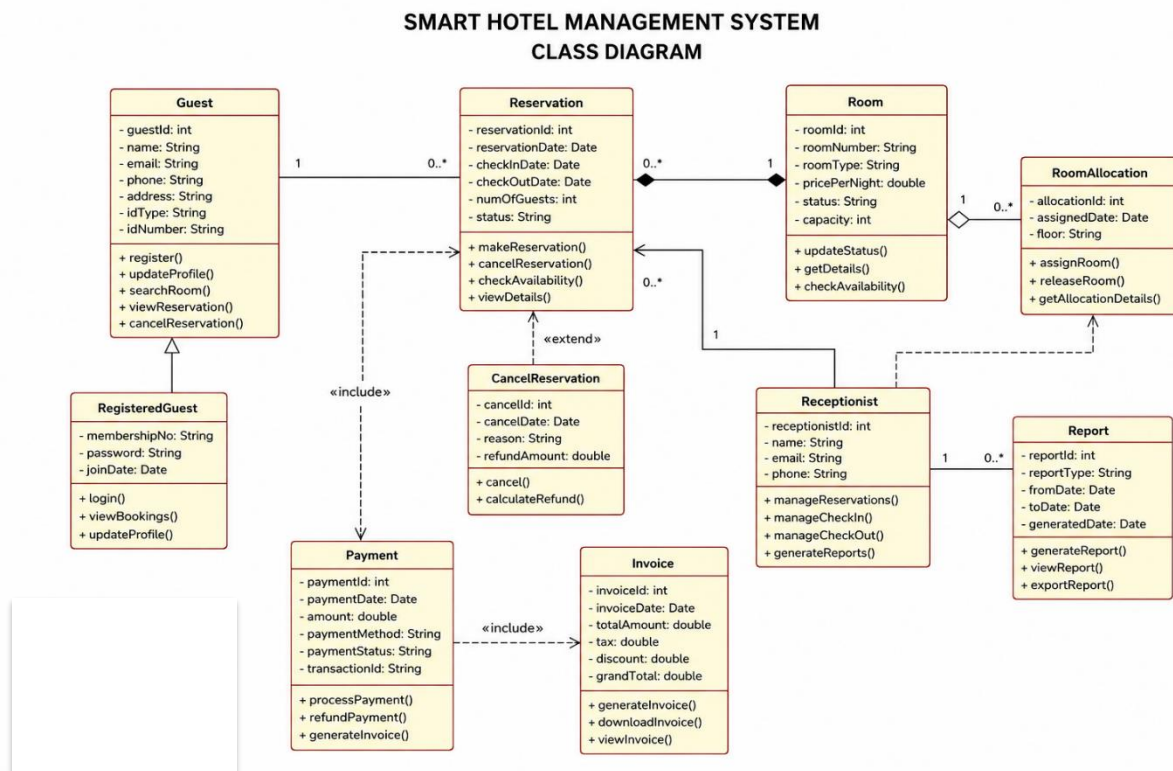
Composition

- Reservation ◆— Payment (**1 : 1**)
- A payment cannot exist without a reservation.

Multiplicity

Class	Relationship	Class	Multiplicity
Guest	books	Reservation	1 : 0..*
Reservation	reserves	Room	1 : 1
Reservation	contains	Payment	1 : 1
Receptionist	manages	Reservation	1 : 0..*

Class Diagram



3. Sequence Diagram Scenario

In an **Online Banking System**, a customer logs in, transfers funds, verifies the transaction using OTP, and receives a transaction confirmation. The communication among the customer, authentication service, banking server, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

Answer:

Classes

1. Guest

Attributes

- guestId : int
- name : String
- phone : String
- email : String

Methods

- register()
- bookRoom()
- cancelReservation()

2. Room

Attributes

- roomId : int
- roomType : String
- roomPrice : double
- status : String

Methods

- checkAvailability()
- updateStatus()

3. Reservation

Attributes

- reservationId : int
- checkInDate : Date
- checkOutDate : Date
- status : String

Methods

- createReservation()
- cancelReservation()
- confirmReservation()

4. Payment

Attributes

- paymentId : int
- amount : double
- paymentMethod : String
- paymentStatus : String

Methods

- makePayment()

- generateReceipt()

5. Receptionist

Attributes

- receptionistId : int
- name : String
- shift : String

Methods

- assignRoom()
- checkInGuest()
- checkOutGuest()

6. Person (Superclass)

Attributes

- id : int
- name : String

Methods

- login()
- logout()

Relationships

Inheritance

- Person <|-- Guest
- Person <|-- Receptionist

Association

- Guest ↔ Reservation (**1** : **0..***)
- Receptionist ↔ Reservation (**1** : **0..***)

Aggregation

- Reservation ◇— Room (**1** : **1**)
- One reservation is associated with one room.

Composition

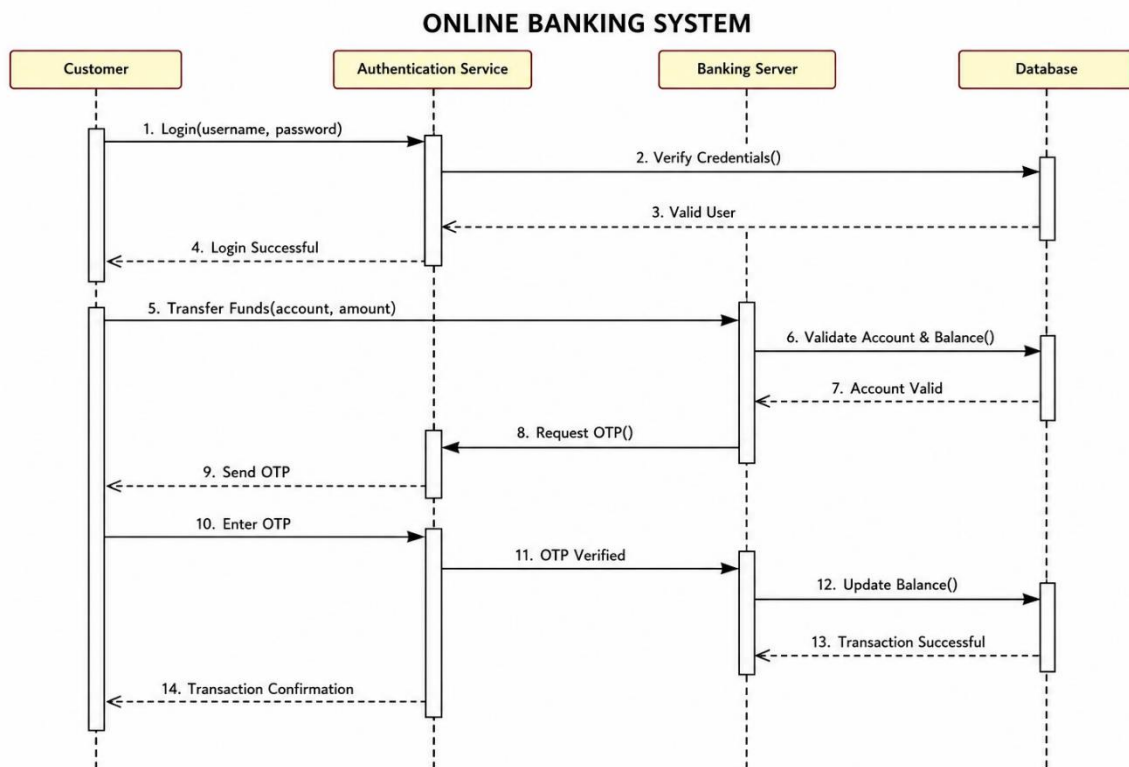
- Reservation ◆— Payment (**1** : **1**)

- A payment cannot exist without a reservation.

Multiplicity

Class	Relationship	Class	Multiplicity
Guest	books	Reservation	1 : 0..*
Reservation	reserves	Room	1 : 1
Reservation	contains	Payment	1 : 1
Receptionist	manages	Reservation	1 : 0..*

Class Diagram



4. Activity Diagram Scenario

A **Passport Application System** allows applicants to submit applications, upload supporting documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

Answer:

Workflow

1. Start
2. Submit Passport Application
3. Upload Supporting Documents
4. **Fork (Parallel Activities)**
 - Identity Verification
 - Processing Fee Payment
5. **Join**
6. Verify Application
7. **Decision Node**
 - **Approved**
 - Print Passport
 - Dispatch Passport
 - Notify Applicant
 - **Rejected**
 - Notify Rejection
8. End

Decision Points

- **Application Approved?**
 - **Yes** → Print and Dispatch Passport
 - **No** → Reject Application and Notify Applicant

Parallel Activities

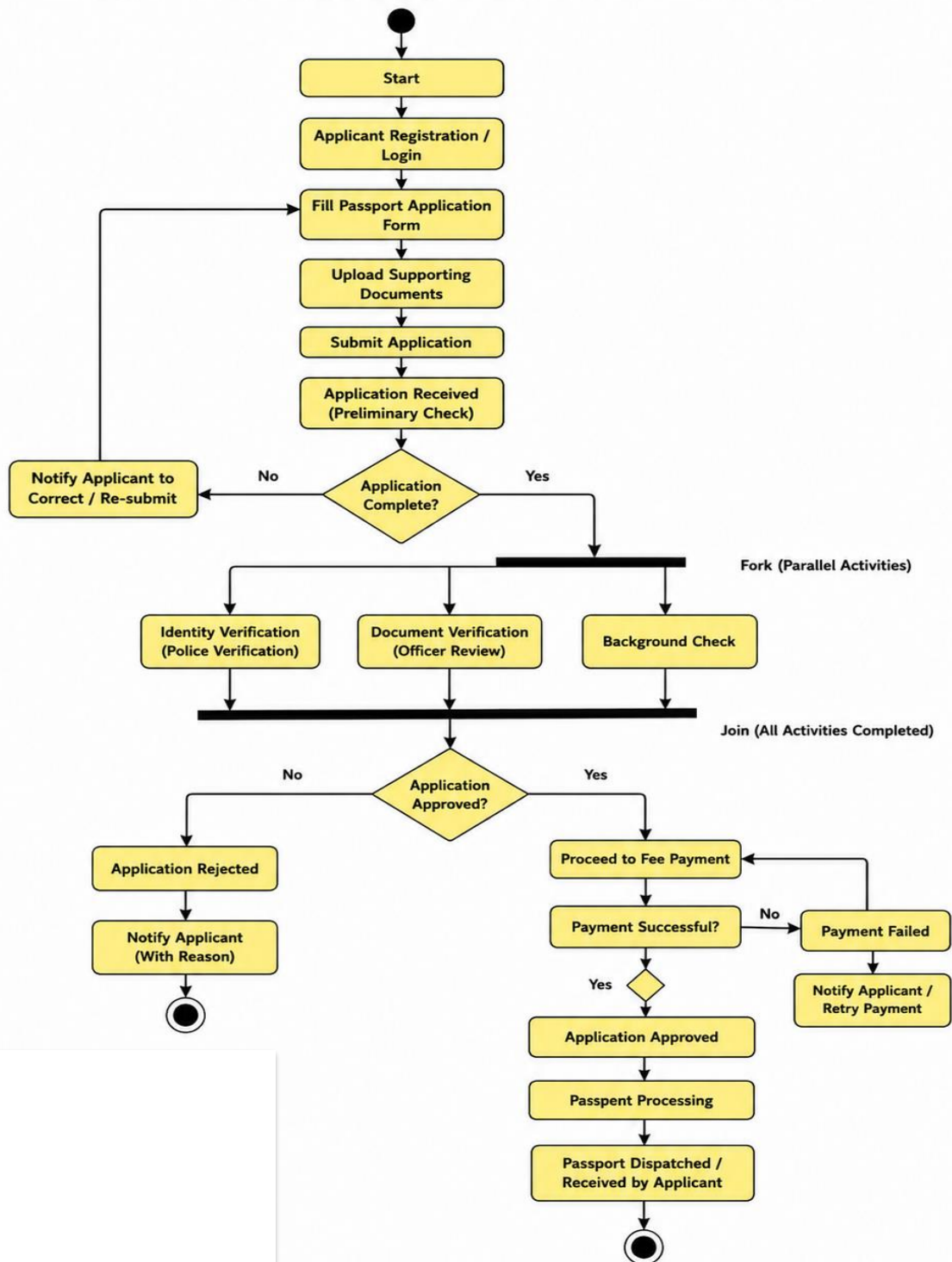
The following activities occur simultaneously:

- Identity Verification
- Processing Fee Payment

These parallel activities are represented using **Fork** and **Join** nodes.

Activity Diagram

PASSPORT APPLICATION SYSTEM – ACTIVITY DIAGRAM



5. State Diagram Scenario

A **Smart Elevator Control System** operates through states such as Idle, Door Open, Door Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The transitions between these states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the elevator's behavior.

Answer:

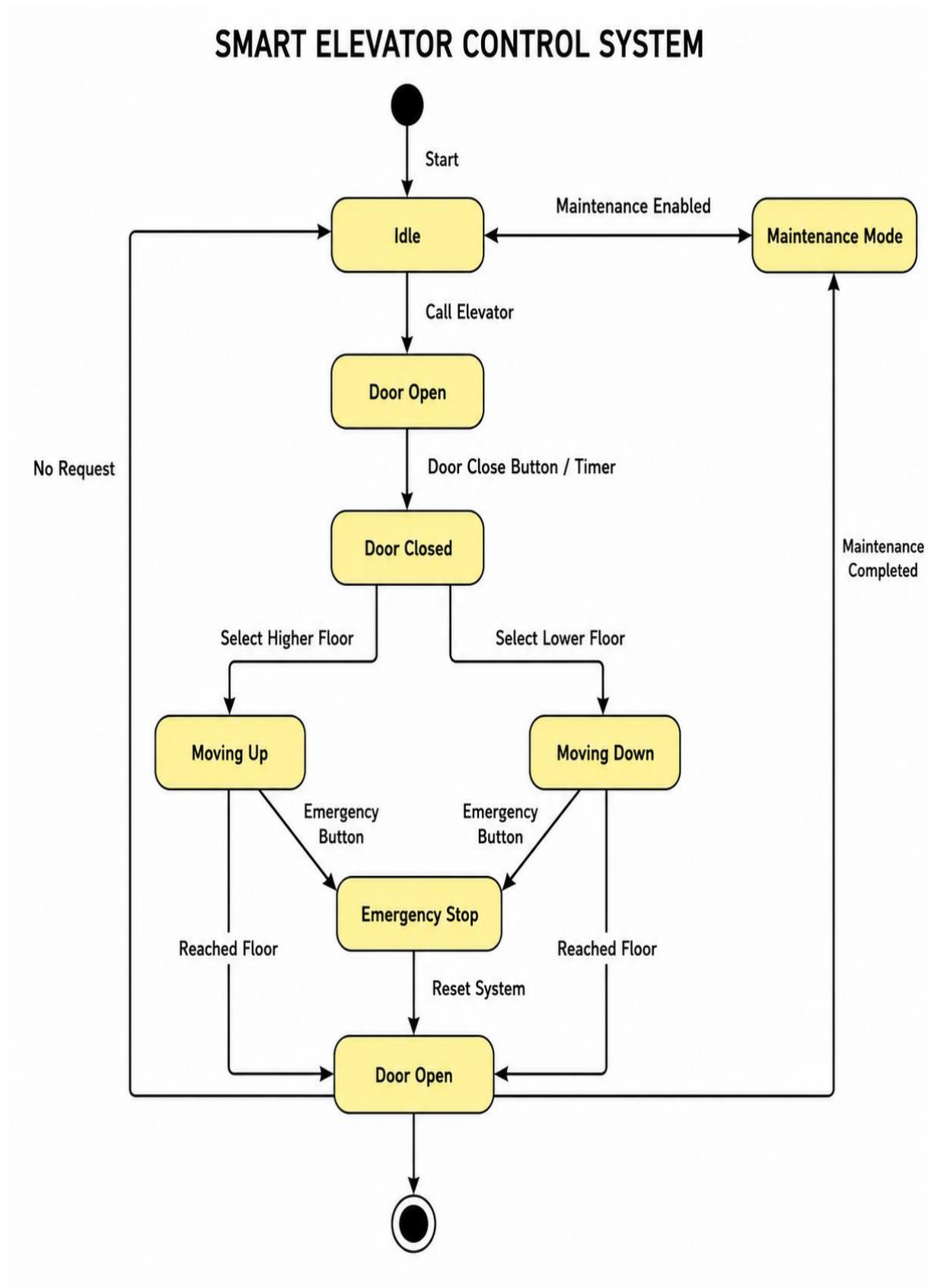
States

- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop
- Maintenance Mode

Triggering Events and Transitions

Current State	Event	Next State
Initial	Power ON	Idle
Idle	Floor Button Pressed	Door Closed
Door Closed	Destination Above	Moving Up
Door Closed	Destination Below	Moving Down
Moving Up	Reached Floor	Door Open
Moving Down	Reached Floor	Door Open
Door Open	Door Close Timer	Idle
Any State	Emergency Button Pressed	Emergency Stop
Emergency Stop	Reset System	Idle
Idle	Maintenance Enabled	Maintenance Mode
Maintenance Mode	Maintenance Completed	Idle

State Diagram



6. Deployment Diagram Scenario

A **Smart Home Automation System** uses smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application to monitor and control lighting, security, and appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

Answer:

Hardware Nodes

- Mobile Device
- Home Gateway
- IoT Controller
- Smart Sensors
- Cloud Server

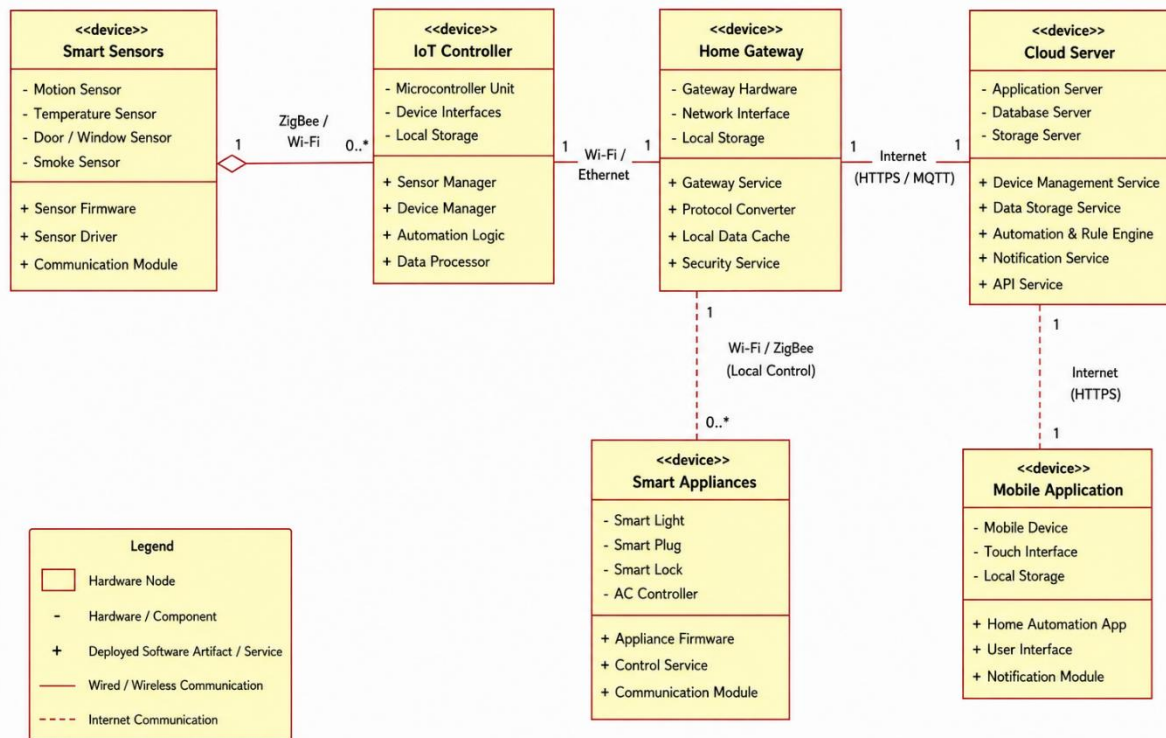
Software Artifacts

- Mobile Application
- Gateway Software
- IoT Control Software
- Sensor Firmware
- Cloud Management System
- Database

Communication Links

- Mobile Device ↔ Cloud Server (Internet)
- Cloud Server ↔ Home Gateway (Internet/Wi-Fi)
- Home Gateway ↔ IoT Controller (Wi-Fi)
- IoT Controller ↔ Smart Sensors (ZigBee/Bluetooth/Wi-Fi)

Deployment Diagram



RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand